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Terminal fitting and connector

Abstract

A terminal fitting **10** is provided with a terminal body portion **11** extending in an axial direction and a plurality of protrusions **20, 30** and **40** projecting in an intersecting direction intersecting the axial direction from the terminal body portion **11**. The plurality of protrusions **20, 30** and **40** are arranged side by side in the axial direction and each protrusion has a top surface **21, 31, 41** along the axial direction, a front surface **22, 32, 42** inclined toward a front side in the axial direction from the top surface **21, 31, 41** and a rear surface **23, 33, 43** inclined toward a rear side in the axial direction from the top surface **21, 31, 41**. Projection amounts of the plurality of protrusions **20, 30** and **40** in the intersecting direction are larger for the protrusions more on the rear side in the axial direction.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4632482	12/1985	Punako	439/744	H01R 13/187
5460549	12/1994	Muzslay	439/733.1	H01R 13/521
6315615	12/2000	Raistrick	439/733.1	H01R 12/585
11658426	12/2022	Txarola	439/744	H01R 12/515
2006/0246786	12/2005	Noguchi	439/751	H01R 12/585
2007/0270001	12/2006	Matsumura	439/82	H05K 3/284
2009/0011658	12/2008	Shibata	439/736	H01R 13/41
2010/0136852	12/2009	Mito	N/A	N/A
2013/0273753	12/2012	Suzuki	439/59	H05K 1/117
2014/0045350	12/2013	Taylor	29/846	H01R 13/24
2015/0064987	12/2014	Nakamura	N/A	N/A
2016/0014118	12/2015	Da et al.	N/A	N/A
2018/0248291	12/2017	Kitajima	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2056410	12/2008	JP	H01R 13/41

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based on and claims priority from Japanese Patent Application No. 2021-214137, filed on Dec. 28, 2021, with the Japan Patent Office, the disclosure of which is incorporated herein in its entirety by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a terminal fitting and a connector.

BACKGROUND

(3) Japanese Patent Laid-open Publication No. 2013-235762 discloses a terminal fitting provided with a terminal body and a press-fitting projection projecting from the terminal body in an orthogonal direction orthogonal to a longitudinal direction of the terminal fitting. The press-fitting

projection includes a front protrusion and a rear protrusion. The press-fitting projection is press-fit into a terminal holding hole of a housing. A terminal fitting having a similar configuration is also disclosed in International Publication No. WO 2009/22604 and Japanese Patent Laid-open Publication Nos. 2018-125225 and 2009-016148.

SUMMARY

(4) A terminal fitting as disclosed in International Publication No. WO 2009/22604 and Japanese Patent Laid-open Publication Nos. 2013-235762, 2018-125225, 2009-016148 and 2016-018595 is required to have a configuration capable of further improving a holding force with the terminal fitting inserted in an insertion hole of a housing.

(5) Accordingly, the present disclosure provides a technique capable of improving a holding force for a terminal fitting.

(6) The present disclosure is directed to a terminal fitting with a terminal body portion extending in an axial direction, and a plurality of protrusions projecting in an intersecting direction intersecting the axial direction from the terminal body portion, the plurality of protrusions being arranged side by side in the axial direction, each protrusion having a top surface along the axial direction, a front surface inclined toward a front side in the axial direction from the top surface and a rear surface inclined toward a rear side in the axial direction from the top surface, and projection amounts of the plurality of protrusions in the intersecting direction being larger for the protrusions more on the rear side in the axial direction.

(7) The present disclosure is directed to a connector with the above terminal fitting and a housing provided with an insertion hole, the terminal fitting being arranged in the insertion hole, the protrusions being locked to the insertion hole.

(8) According to the present disclosure, a holding force of a terminal fitting can be improved.

(9) The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, and features will become apparent by reference to the drawings and the following detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a perspective view showing a terminal fitting.

(2) FIG. **2** is a plan view showing the terminal fitting.

(3) FIG. **3** is a side view showing the terminal fitting.

(4) FIG. **4** is an enlarged view of the terminal fitting viewed from a tip side.

(5) FIG. **5** is a section of an insertion hole and the periphery thereof before the insertion of the terminal fitting in a housing.

(6) FIG. **6** is a section of the insertion hole and the periphery thereof in a state where the terminal fitting is inserted in the insertion hole of the housing.

(7) FIG. **7** is an enlarged view of the terminal fitting viewed from the tip side showing the state where the terminal fitting is inserted in the insertion hole of the housing.

(8) FIG. **8** is a perspective view showing a connector.

DETAILED DESCRIPTION

(9) In the following detailed description, reference is made to the accompanying drawings, which form a part hereof. The illustrative embodiments described in the detailed description, drawings, and claims are not meant to be limiting. Other embodiments may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter presented here.

DESCRIPTION OF EMBODIMENTS OF PRESENT DISCLOSURE

(10) First, embodiments of the present disclosure are listed and described.

(11) (1) The terminal fitting of the present disclosure is provided with a terminal body portion extending in an axial direction, and a plurality of protrusions projecting in an intersecting direction intersecting the axial direction from the terminal body portion, the plurality of protrusions being arranged side by side in the axial direction, each protrusion having a top surface along the axial direction, a front surface inclined toward a front side in the axial direction from the top surface and a rear surface inclined toward a rear side in the axial direction from the top surface, and projection amounts of the plurality of protrusions in the intersecting direction being larger for the protrusions more on the rear side in the axial direction.
(12) According to the configuration of the present disclosure, the terminal fitting is inserted into an insertion hole of a connector housing from the front side in the axial direction and, when the plurality of protrusions interfere with a wall portion of the insertion hole, the wall portion deformed by being pushed and expanded by the protrusions enters behind the rear surfaces of the protrusions. Besides, since the projection amounts in the intersecting direction are larger for the protrusions more on the rear side in the axial direction, the wall portion deformed by being pushed and expanded by the protrusion on the front side in the axial direction is easily locked by the protrusion on the rear side in the axial direction. Further, since the top surfaces of the protrusions are configured along the axial direction, contact surfaces with the wall portion of the insertion hole are easily secured. Therefore, a force for holding the terminal fitting in the connector housing can be improved.
(13) (2) Preferably, joining portions are provided by rounding a connected part of the top surface and the front surface and a connected part of the top surface and the rear surface.
(14) According to this configuration, the wall portion of the insertion hole is easily deformed by being smoothly pushed and expanded by the protrusion and the deformation of the wall portion of the insertion hole pushed and expanded by the protrusion is easily restored after the passage of the protrusion as compared to a configuration in which the connected part of the top surface and the front surface and the connected part of the top surface and the rear surface are angular.
(15) (3) Preferably, angles of inclination of the front surfaces in the plurality of protrusions are larger for the protrusions more on the rear side in the axial direction.
(16) According to this configuration, since the wall portion of the insertion hole is first pushed and expanded by the front surface of the protrusion having a relatively small angle of inclination, out of the plurality of protrusions, insertion resistance is relatively small and the terminal fitting is easily smoothly inserted. In the protrusion having a relatively large projection amount in the intersecting direction, the wall portion of the insertion hole is pushed and expanded relatively more, but the wall portion of the insertion hole is easily pushed and expanded since the angle of inclination of the front surface is relatively large.
(17) (4) Preferably, a bottom surface along the axial direction is provided behind the rear surface of the protrusion in the terminal body portion.
(18) According to this configuration, in the terminal body portion, the bottom surface capable of securing a length in the axial direction can be arranged on the terminal body portion, and the wall portion of the insertion hole easily enters behind the rear surface of the protrusion.
(19) The connector of the present disclosure is provided with the terminal fitting of any one of (1) to (4) and a housing provided with an insertion hole, the terminal fitting being arranged in the insertion hole, the protrusions being locked to a wall portion of the insertion hole.
(20) According to this configuration, a connector achieving effects similar to those of the above terminal fitting can be realized.
Details of Embodiment of Present Disclosure
Embodiment
(21) A specific embodiment of a terminal fitting of the present disclosure is described below with reference to FIGS. **1** to **8**. In this embodiment, an axial direction of a terminal fitting **10** is a front-rear direction. A tip side in the axial direction of the terminal fitting **10** is a front side (side

indicated by an arrow F shown in FIGS. **2** and **6**), and a side opposite to the tip side is a rear side (side indicated by an arrow R shown in FIGS. **2** and **6**).

Configuration of Terminal Fitting

(22) The terminal fitting **10** of this embodiment is a so-called male terminal as shown in FIG. **1**. The terminal fitting **10** is formed by stamping an electrically conductive metal plate, for example, made of copper, copper alloy or the like by press-working. Plating may be applied to the surface of the terminal fitting **10**.

(23) As shown in FIGS. **1** to **3**, the terminal fitting **10** is in the form of a solid rectangular bar elongated in the axial direction (front-rear direction) as a whole. The terminal fitting **10** includes a terminal body portion **11** in the form of a solid rectangular bar extending in the axial direction. The terminal body portion **11** includes a tab **12** in a front part and a board connecting portion **13** in a rear part. The board connecting portion **13** is inserted into a connection hole (not shown) of a printed circuit board and electrically connected to an electrically conductive portion of the printed circuit board. Note that the board connecting portion **13** is bendable if necessary.

(24) In the terminal body portion **11**, a press-fitting portion **14** and a stopper portion **15** are provided between the tab **12** and the board connecting portion **13**. The press-fitting portion **14** is press-fit into an insertion hole **51** provided in a housing **50** (see FIGS. **5** to **7**). The housing **50** is, for example, made of resin. The press-fitting portion **14** includes a plurality of protrusions **20**, **30** and **40**. The protrusions **20**, **30** and **40** project in an intersecting direction intersecting the axial direction (front-rear direction) from the terminal body portion **11**. The protrusions **20**, **30** and **40** are provided on each of both sides in the intersecting direction of the terminal body portion **11**. The respective protrusions **20**, **30** and **40** are arranged side by side in the axial direction like saw teeth. As shown in FIG. **6**, if the press-fitting portion **14** is press-fit into the insertion hole **51**, the respective protrusions **20**, **30** and **40** are locked by biting into a wall portion **52** of the insertion hole **51** and the terminal fitting **10** is retained and held in the housing **50**. The board connecting portion **13** is exposed outside the housing **50**.

(25) As shown in FIG. **1**, the stopper portion **15** is located behind the press-fitting portion **14** and provided with a pair of protruding portions **16**. As shown in FIG. **6**, by inserting the press-fitting portion **14** into the insertion hole **51** from behind and stopping the pair of protruding portions **16** in contact with a rear surface **53** of the housing **50**, a mounted position of the terminal fitting **10** in the housing **50** is specified. With the terminal fitting **10** mounted in the housing **50**, the tab **12** is arranged to project into a receptacle (not shown) of the housing **50**.

(26) Both upper and lower surfaces of the terminal body portion **11**, the press-fitting portion **14** and the stopper portion **15** shown in FIG. **3** are flat along the axial direction and arranged in parallel to each other.

(27) Side surfaces (surfaces along a vertical direction shown in FIG. **4**) of the tab **12** are both formed in parallel to the vertical direction shown in FIG. **4**. Four corner parts of the tab **12** are chamfered. The tip (front end) of the tab **21** is formed in a tapered manner to reduce dimensions. The terminal fitting **10** can be easily press-fit into the insertion hole **51** of the housing **50** by the tapered tip.

Configuration of Press-Fitting Portion

(28) The plurality of protrusions **20**, **30** and **40** are also called first protrusions **20**, second protrusions **30** and third protrusions **40** in an order of arrangement from the tip side (front side) to the opposite side (rear side) in the axial direction (front-rear direction). Adjacent ones of the first, second and third protrusions **20**, **30** and **40** are arranged apart from each other. Note that the first, second and third protrusions **20**, **30** and **40** on one side (upper side of FIG. **2**) in the intersecting direction of the press-fitting portion **14** are described below, but the first, second and third protrusions **20**, **30** and **40** on the other side (lower side of FIG. **2**) in the intersecting direction are also similarly configured.

(29) As shown in FIG. **2**, the first protrusion **20** is provided with a top surface **21**, a front surface **22**

and a rear surface **23**. The top surface **21** is along the axial direction. More specifically, the top surface **21** is a flat surface orthogonal to the intersecting direction. The front surface **22** is inclined toward a front side in the axial direction from the top surface **21**. That is, the front surface **22** is inclined downward toward the front side in the axial direction. The rear surface **23** is inclined toward a rear side in the axial direction from the top surface **21**. That is, the rear surface **23** is inclined downward toward the rear side in the axial direction.

(30) As shown in FIG. **2**, the second protrusion **30** is also provided with a top surface **31**, a front surface **32** and a rear surface **33** configured similarly to the top surface **21**, the front surface **22** and the rear surface **23** of the first protrusion **20**. The third protrusion **40** is also provided with a top surface **41**, a front surface **42** and a rear surface **43** configured similarly to the top surface **21**, the front surface **22** and the rear surface **23** of the first protrusion **20**.

(31) As shown in FIG. **6**, when the terminal fitting **10** is inserted into the insertion hole **51** of the housing **50** from the front side in the axial direction, the plurality of protrusions **20**, **30** and **40** interfere with the wall portion **52** of the insertion hole **51**. Here, an inserting direction of the terminal fitting **10** into the insertion hole **51** is a direction along the axial direction. The wall portion **52** deformed by being pushed and expanded by the protrusions **20**, **30** and **40** enters clearances between adjacent ones of the protrusions (more specifically, behind the rear surfaces **23**, **33** and **43** of the respective protrusions **20**, **30** and **40**). The wall portion **53** having entered behind the rear surfaces **23** of the first protrusions **20** is locked by the front surfaces **32** of the second protrusions **30**. The wall portion **53** having entered behind the rear surfaces **33** of the second protrusions **30** is locked by the front surfaces **42** of the third protrusions **40**.

(32) Since the respective top surfaces **21**, **31** and **41** extend along the axial direction, contact surfaces with the wall portion **52** of the insertion hole **51** are easily secured. Further, since the tips of the respective protrusions **20**, **30** and **40** are not pointed, a load to the wall portion **52** of the insertion hole **51** can be dispersed and the breakage and scraping of the wall portion **52** during insertion and during holding can be suppressed. Since the respective front surfaces **22**, **32** and **42** are inclined toward the front side in the axial direction, the wall portion **52** of the insertion hole **51** can be smoothly pushed and expanded while being hardly scratched. Since the rear surface **23** is inclined toward the rear side in the axial direction, the part of the wall portion **52** of the insertion hole **51** entering behind the rear surface **23** is easily pressed against the front surface **32** of the second protrusion **30** located behind. Since the rear surface **33** is inclined toward the rear side in the axial direction, the part of the wall portion **52** of the insertion hole **51** entering behind the rear surface **33** is easily pressed against the front surface **42** of the third protrusion **40** located behind. Therefore, a force for holding the terminal fitting **10** in the housing **50** can be improved.

(33) As shown in FIG. **2**, projection amounts of the plurality of protrusions **20**, **30** and **40** in the intersecting direction are larger for the protrusions located more on the rear side in the axial direction. Specifically, if H1 denotes a projection amount of the first protrusion **20** in an orthogonal direction orthogonal to the axial direction, H2 denotes a projection amount of the second protrusion **30** in the orthogonal direction and H3 denotes a projection amount of the third protrusion **40** in the orthogonal direction, there is a relationship of $H1<H2<H3$. Since the projection amounts in the intersecting direction are larger for the protrusions more on the rear side in the axial direction as just described, a locking margin of the wall portion **52** of the insertion hole **51** deformed by being pushed and expanded by the protrusion on the front side in the axial direction and the protrusion located on the rear side in the axial direction is easily secured. Therefore, the force for holding the terminal fitting **10** in the housing **50** can be improved.

(34) As shown in FIG. **2**, a joining portion **24** is provided by rounding a connected part of the top surface **21** and the front surface **22** in the first protrusion **20**. The joining portion **24** is convexly curved outward in the intersecting direction (side opposite to the terminal body portion **11**). As compared to a configuration in which the connected part of the top surface **21** and the front surface **22** is angular, the wall portion **52** of the insertion hole **51** is easily deformed by being smoothly

pushed and expanded by the first protrusion **20**. A joining portion **25** is provided by rounding a connected part of the top surface **21** and the rear surface **23** in the first protrusion **20**. The joining portion **24** is convexly curved outward in the intersecting direction (side opposite to the terminal body portion **11**). As compared to a configuration in which the connected part of the top surface **21** and the rear surface **23** is angular, the deformation of the wall portion **52** of the insertion hole **51** pushed and expanded by the first protrusion **20** is easily restored after the passage of the first protrusion **20**.

(35) As shown in FIG. **2**, joining portions **34**, **35** configured similarly to the projections **24**, **25** of the first protrusion **20** are provided also on the second protrusion **30**. Jointing portions **44**, **45** configured similarly to the projections **24**, **25** of the first protrusion **20** are provided also on the third protrusion **40**.

(36) As shown in FIG. **2**, angles of inclination (angles of inclination with respect to the axial direction) of the front surfaces of the plurality of protrusions **20**, **30** and **40** are larger for the protrusions more on the rear side in the axial direction. Specifically, if $\theta 1$ denotes the angle of inclination of the front surface **22** of the first protrusion **20**, $\theta 2$ denotes the angle of inclination of the front surface **32** of the second protrusion **30** and $\theta 3$ denotes the angle of inclination of the front surface **42** of the third protrusion **40**, there is a relationship of $\theta 1<\theta 2<\theta 3$. Since the wall portion **52** of the insertion hole **51** is first pushed and expanded by the front surfaces **22** of the first protrusion **20** having a relatively small angle of inclination, out of the plurality of protrusions **20**, **30** and **40**, insertion resistance is relatively small and the terminal fitting **10** is easily smoothly inserted. The wall portion **52** of the insertion hole **51** is pushed and expanded relatively more by the protrusions **30**, **40** having a relatively large projection amount in the intersecting direction, but the wall portion **52** of the insertion hole **51** is easily pushed and expanded since the angles of inclination of the front surfaces **32**, **42** are relatively large.

(37) As shown in FIG. **2**, angles of inclination (angles of inclination with respect to the axial direction) of the rear surfaces of the plurality of protrusions **20**, **30** and **40** are larger for the protrusions more on the rear side in the axial direction. Specifically, if $\theta 4$ denotes the angle of inclination of the rear surface **23** of the first protrusion **20**, $\theta 5$ denotes the angle of inclination of the rear surface **33** of the second protrusion **30** and $\theta 6$ denotes the angle of inclination of the rear surface **43** of the third protrusion **40**, there is a relationship of $\theta 4<\theta 5<\theta 6$.

(38) As shown in FIG. **2**, a bottom surface **26** is provided behind the rear surface **23** of the first protrusion **20** in the terminal body portion **11**. The bottom surface **26** extends along the axial direction. More specifically, the bottom surface **26** is a flat surface orthogonal to the intersecting direction. The bottom surface **26** capable of ensuring a length in the axial direction can be arranged behind the rear surface **23** of the first protrusion **20**, and the wall portion **52** of the insertion hole **51** more easily enters behind the rear surface **23**. In the terminal body portion **11**, a bottom surface **36** configured similarly to the bottom surface **26** is provided also behind the rear surface **33** of the second protrusion **30**. In the terminal body portion **11**, a bottom surface **46** configured similarly to the bottom surface **26** is provided also behind the rear surface **43** of the third protrusion **40**.

(39) As shown in FIG. **2**, the lengths along the axial direction are larger for the bottom surfaces located more on the rear side in the axial direction. Specifically, if L1 denotes the length along the axial direction of the bottom surface **26**, L2 denotes the length along the axial direction of the bottom surface **36** and L3 denotes the length along the axial direction of the bottom surface **46**, there is a relationship of $L1<L2<L3$. In this way, the deformed wall portion **52** of the insertion hole **51** more easily approaches the bottom surfaces more on the rear side.

(40) As shown in FIG. **2**, a connected part of the rear surface **23** of the first protrusion **20** and the bottom surface **26** is in the form of a curved surface. Similarly, a connected part of the rear surface **33** of the second protrusion **30** and the bottom surface **36** is in the form of a curved surface. Similarly, a connected part of the rear surface **43** of the third protrusion **40** and the bottom surface **46** is in the form of a curved surface.

(41) As shown in FIG. **2**, a connected part of the bottom surface **26** and the front surface **32** of the second protrusion **30** is in the form of a curved surface. Similarly, a connected part of the bottom surface **36** and the front surface **42** of the third protrusion **40** is in the form of a curved surface.

Configuration of Connector

(42) A connector **100** of this embodiment is provided with the terminal fittings **10** and the housing **50** as shown in FIG. **8**. The housing **50** is provided with the insertion holes **51** in which the terminal fittings **10** are arranged. As shown in FIG. **6**, the protrusions **20**, **30** and **40** are locked to the wall portion **52** of the insertion hole **51**.

Effects of Embodiment

(43) In the terminal fitting **10** of this embodiment, the plurality of protrusions are arranged side by side in the axial direction and each has the top surface along the axial direction, the front surface inclined toward the front side in axial direction from the top surface and the rear surface inclined toward the rear side in the axial direction from the top surface, and the projection amounts in the intersecting direction of the plurality of protrusions are larger for the protrusions more on the rear side in the axial direction.

(44) According to the configuration of the present disclosure, the terminal fitting **10** is inserted into the insertion hole **51** of the housing **50** from the front side in the axial direction and, when the plurality of protrusions **20**, **30** and **40** interfere with the wall portion **52** of the insertion hole **51**, the wall portion **52** deformed by being pushed and expanded by the protrusions **20**, **30** and **40** enters behind the rear surfaces **23**, **33** and **43** of the protrusions **20**, **30** and **40**. Besides, since the projection amounts in the intersecting direction are larger for the protrusions more on the rear side in the axial direction, the wall portion **52** deformed by being pushed and expanded by the protrusion on the front side in the axial direction is easily locked by the protrusion on the rear side in the axial direction. Further, since the top surfaces **21**, **31** and **41** of the protrusions **20**, **30** and **40** are configured along the axial direction, contact surfaces with the wall portion **52** of the insertion hole **51** are easily secured. Therefore, a force for holding the terminal fitting **10** in the housing **50** can be improved.

(45) Further, in the first protrusion **20**, the joining portions **24**, **25** are provided by rounding the connected part of the top surface **21** and the front surface **22** and the connected part of the top surface **21** and the rear surface **23**. The wall portion **52** of the insertion hole **51** is easily deformed by being smoothly pushed and expanded by the first protrusion **20** as compared to a configuration in which the connected part of the top surface **21** and the front surface **22** is angular. The deformation of the wall portion **52** of the insertion hole **51** pushed and expanded by the first protrusion **20** is easily restored after the passage of the first protrusion **20** as compared to a configuration in which the connected part of the top surface **21** and the rear surface **23** is angular. Similar effects are achieved also for the second and third protrusions **30**, **40**.

(46) Further, the angles of inclination of the front surfaces in the plurality of protrusions **20**, **30** and **40** are larger for the protrusions more on the rear side in the axial direction. Thus, the wall portion **52** of the insertion hole **51** is first pushed and expanded by the front surfaces **22** of the first protrusions **20** having a relatively small angle of inclination, out of the plurality of protrusions **20**, **30** and **40**, wherefore insertion resistance is relatively small and the terminal fitting **10** is easily smoothly inserted. In the second and third protrusions **30**, **40** having a relatively large projection amount in the intersecting direction, the wall portion **52** of the insertion hole **51** is pushed and expanded relatively more, but the wall portion **52** of the insertion hole **51** is easily pushed and expanded since the angles of inclination of the front surfaces **32**, **42** are relatively large.

(47) Further, the bottom surfaces **26**, **36** and **46** along the axial direction are provided behind the rear surfaces **23**, **33** and **43** of the protrusions **20**, **30** and **40** in the terminal body portion **11**. According to this configuration, the bottom surfaces **26**, **36** and **46** capable of ensuring lengths in the axial direction can be arranged in the terminal body portion **11** and the wall portion **52** of the insertion hole **51** more easily enters behind the rear surfaces **23**, **33** and **43** of the protrusions **20**, **30**

and **40**.
Other Embodiments
(48) The present invention is not limited to the above described and illustrated embodiment and is represented by claims. The present invention is intended to include all changes in the scope of claims and in the meaning and scope of equivalents and also include the following embodiment.
(49) Although the terminal fitting **10** includes three protrusions **20**, **30** and **40** in the above embodiment, two, four or more protrusions may be provided.
(50) Although the terminal body portion **11** is in the form of a rectangular bar in the above embodiment, the terminal body portion **11** may have another shape such as a cylindrical shape.
(51) In the above embodiment, the projection amount of the protrusion in the intersecting direction may be equal between the first and second protrusions **20**, **30** or may be equal between the second and third protrusions **30**, **40**.
(52) In the above embodiment, the angle of inclination of the front surface of the protrusion may be equal between the first and second protrusions **20**, **30** or may be equal between the second and third protrusions **30**, **40**.
(53) From the foregoing, it will be appreciated that various exemplary embodiments of the present disclosure have been described herein for purposes of illustration, and that various modifications may be made without departing from the scope and spirit of the present disclosure. Accordingly, the various exemplary embodiments disclosed herein are not intended to be limiting, with the true scope and spirit being indicated by the following claims.

Claims

1. A terminal fitting, comprising: a terminal body portion extending in an axial direction; and a plurality of protrusions projecting in an intersecting direction intersecting the axial direction from the terminal body portion, the plurality of protrusions being arranged side by side in the axial direction, each protrusion having a top surface along the axial direction, a front surface inclined frontward in the axial direction from the top surface and a rear surface inclined rearward in the axial direction from the top surface, wherein projection amounts of the plurality of protrusions in the intersecting direction increase as the protrusions are positioned further rearward along the axial direction, the terminal body portion includes bottom surfaces along the axial direction, each positioned between the rear surface of one of the plurality of protrusions and the front surface of another of the plurality of protrusions, and lengths of the bottom surfaces in the axial direction increase as the bottom surfaces are positioned further rearward along the axial direction.
2. The terminal fitting of claim 1, wherein joining portions are provided by rounding a connected part of the top surface and the front surface and a connected part of the top surface and the rear surface.
3. The terminal fitting of claim 1, wherein angles of inclination of the front surfaces in the plurality of protrusions increase as the protrusions are positioned further rearward along the axial direction.
4. A connector, comprising: the terminal fitting of claim 1; and a housing provided with an insertion hole, the terminal fitting being arranged in the insertion hole, the protrusions being locked to a wall portion of the insertion hole.
5. The connector of claim 4, wherein the terminal fitting further includes a stopper provided on the terminal body portion between the plurality of protrusions and a board connecting portion on a rear side of the terminal body portion, and the stopper is configured to contact an outer rear surface of the housing to stop insertion of the terminal body portion into the insertion hole.